

KYLE THOMPSON

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EDUCATION

Northwestern University

Expected Dec 2026

Master of Science in Robotics

University of Illinois at Urbana-Champaign

Aug 2020 - Dec 2024

Bachelor of Science in Mechanical Engineering, Minor in Computer Science

TECHNICAL SKILLS

Software Development: C, C++, Python, Java, MATLAB/Simulink, Bash, Git, Linux, Docker, Unit Testing, CI/CD, Pandas

Robotics: ROS 2, OpenCV, Motion Planning, Inverse Kinematics, MoveIt2, SLAM, Nav2, EKF, MPC, Camera Calibration

Simulation & ML: PyTorch, World Models, RL, Gazebo, Isaac Sim, MuJoCo, VAE, YOLO, CLIP, SAM, HuggingFace

Hardware: LiDAR, IMUs, RGB-D, NVIDIA Jetson, Zephyr RTOS, UART, 3D Printing, Rapid Prototyping, NX, SolidWorks, DDS

WORK EXPERIENCE

Vermeer

Brooklyn, New York

Robotics Software Engineering Intern

Jun 2026 – Sep 2026

- Optimized FR-20 cobot EO/IR camera calibration, cutting cycle time from 15 to 8 min; built pytest integration tests
- Integrated end-effector trajectory spline to reduce camera-IMU reprojection error 40% with 20% more coverage
- Root-caused intermittent calibration failures from logs, added regression tests and doubled calibration success rate
- Deployed containerized machine learning inference on NVIDIA Jetson Orin for EKF-SLAM covariance scaling analysis

Willow

Mountain View, CA

Engineering Intern

Apr 2025 – Aug 2025

- Prototyped multi-antenna Bluetooth Channel Sounding distance measurement embedded firmware with C
- Engineered Zephyr RTOS, UART communication protocols, multithread synchronization in proof-of-concept system
- Performed DFU and system validation of embedded firmware on 150 Willow Sync pumps

Mechanical Engineering Intern

Jun 2024 – Aug 2024

- Applied NX and SLA printing to rapidly fabricate soft plastic prototypes in design sprint to product release
- Created test models and generated vacuum waveforms to improve prototype system efficiency by 20%

Ancora Heart

Santa Clara, CA

R&D Intern

Aug 2023 – Dec 2023

- Executed verification testing, fixture development and data analysis to support catheter performance evaluation
- Assembled, programmed C, and installed automated catheter braiding system to reduce braiding time by 50%

PROJECT EXPERIENCE

Robot- and Task-Agnostic Data Driven World Model (ROS 2, PyTorch)

Mar 2026 – Current

- Integrated CEM-MPC and RSSM World Model on ROS 2 to learn environment dynamics from camera images only
- Implemented a reward model and image-goal conditioning in architecture to guide tasks in MuJoCo simulation
- Developed velocity IK controllers and CBF safety filter for autonomous data collection on Stretch 3 and WidowX-200

Language Guided SLAM on Unitree Go2 (ROS 2, Python, C++)

Jan 2026 – Mar 2026

- Built a 3D semantic concept map fusing SAM3, CLIP, and RGB-D depth, queryable by language to set Nav2 goals
- Benchmarked 5 SAM3/CLIP/GroundingDINO pipelines on a remote GPU to select 0.73 IoU SAM3+CLIP for mapping
- Applied RTAB-Map, Nav2, sensor fusion, and nearest-frontiers on Go2 for SLAM and autonomous exploration
- Resolved multi-machine ROS 2 networking issues utilizing CycloneDDS discovery, QoS tuning, and TF sync restamping

Simultaneous Localization and Mapping (SLAM) from Scratch (ROS 2, C++)

Jan 2026 – Mar 2026

- Engineered Extended Kalman Filter SLAM predict/correct cycle in C++ for TurtleBot3 using ROS 2 and Armadillo
- Implemented a landmark detection algorithm from LiDAR clustering using data association for state estimation
- Leveraged CMake and Catch2 to build and validate a modular navigation stack with comprehensive unit testing

Ball Catching Franka Arm (Python, ROS 2, YOLO)

Nov 2025 – Dec 2025

- Led team of 4 engineers to develop motion planning and collision avoidance MoveIt2 ROS 2 package for Franka arm
- Trained YOLOv11 on 5,000 images for ball tracking and trajectory regression, intercepting throws within 1 s of flight
- Conducted eye-in-hand camera calibration using RGB-D camera and ArUco markers for precise 3D localization